LABORATORY · Lab · 26 hours

Experiment 8: DC Position Control System Using MATLAB/Simulink



Control Systems Laboratory (22EEG53)

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Learning objectives

- Model a DC motor position control system
- Obtain its step response for a given controller gain
- Observe the effect of gain on overshoot and settling
- Understand the role of feedback

Syllabus context

Laboratory component of 22EEG53 Control Systems (0:0:2 hours per week). CIE: record and observation 25, viva-voce 5, lab test 20 marks.

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Aim and requirements

Aim: to simulate a DC position control system in MATLAB/Simulink and obtain its step response.

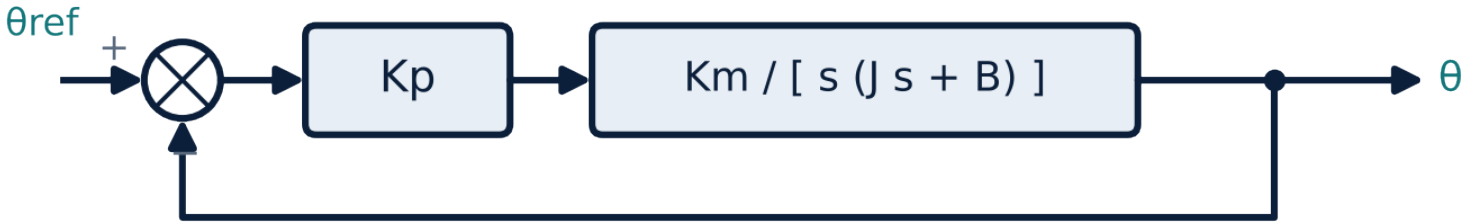
Software: MATLAB with the Control System Toolbox (Simulink where required).

Hardware: Personal computer.

Motor data (example): Ra = 1 Ω, La = 0.5 H, J = 0.01 kg·m², B = 0.1 N·m·s, Kt = Kb = 0.01.

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Block diagram



Unity feedback position loop with proportional controller Kp

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Motor transfer function

$$\theta(s)/Va(s) = Kt / \{ s [(La\ s + Ra)(J\ s + B) + Kt\ Kb] \}$$

angle / armature voltage

$$\text{Closed loop} = Kp\ G/(1 + Kp\ G)$$

unity feedback

- Type 1 system: zero steady-state error for a step.
- Increasing Kp reduces damping and increases overshoot.

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Procedure

1

Parameters

Enter Ra, La, J, B, Kt, Kb in the workspace.

2

Plant

G = tf(Kt, [La*J, La*B+Ra*J, Ra*B+Kt*Kb, 0]).

3

Closed loop

Add Kp and unity feedback.

4

Simulate

Step response for several Kp; read Mp, ts.

5

Simulink

Step → Sum → Gain → Transfer Fcn → Scope, with unity feedback.

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MATLAB program

```
Ra=1; La=0.5; J=0.01; B=0.1; Kt=0.01; Kb=0.01;
G = tf(Kt, [La*J, La*B+Ra*J, Ra*B+Kt*Kb, 0]);
Kp = [1 5 20 50];  hold on
for k = 1:length(Kp)
  step(feedback(Kp(k)*G, 1), 0:0.05:20)
end
legend('Kp=1','5','20','50'); grid on
stepinfo(feedback(20*G,1))
```

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Observation table

Kp	Mp (%)	ts (s)	Steady-state error
1			
5			
20			
50			

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Key takeaways

- DC motor: $\theta/Va = Kt/\{s[(Las + Ra)(Js + B) + KtKb]\}$
- Position loop is type 1: zero step error
- Higher Kp: faster but more overshoot
- Simulink and MATLAB give the same result

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Practice questions

Q1. Derive the DC motor transfer function.

Q2. Why is the steady-state error zero for a step input?

Q3. What is the effect of Kp on damping?

Q4. Name the blocks needed in Simulink.